AI-Based Classroom Photo Attendance Management System: CampusEYE

1st Md. Arebi Sarker
Dept. of Electrical & Computer Engineering
North South University
Dhaka, Bangladesh
arebi.sarker@northsouth.edu

2nd Md. Rizvi Ahmed
Dept. of Electrical & Computer Engineering
North South University
Dhaka, Bangladesh
rizvi.safin@northsouth.edu

3rd Safayat Ibrahim
Dept. of Electrical & Computer Engineering
North South University
Dhaka, Bangladesh
safayat.ibrahim@northsouth.edu

4th Rafiqul Islam
Dept. of Electrical & Computer Engineering
North South University
Dhaka, Bangladesh
rafiqul.islam12@northsouth.edu

***Abstract*—CampusEYE is an AI-powered classroom attendance system that utilizes facial recognition to automate attendance management. An instructor captures a single classroom photograph, from which the system identifies students and records attendance automatically. The proposed approach aims to reduce manual workload, human error, and proxy attendance while improving classroom management efficiency.**

***Index Terms*—Face Recognition, Attendance Management, ArcFace, RetinaFace, Computer Vision**

I. Introduction

Traditional attendance methods, including manual roll calls, attendance sheets, RFID cards, and fingerprint scanners, are often time-consuming, error-prone, and susceptible to proxy attendance. CampusEYE addresses these limitations through an AI-based system that identifies multiple students from a single classroom photograph and automatically records their attendance.

II. Project Overview

CampusEYE combines computer vision, deep learning, and modern web technologies. A teacher captures a classroom image using a mobile phone or web application and uploads it securely through a REST API. The AI engine identifies the students present and stores attendance records in a database. Faculty members can review uncertain matches, correct records, and generate attendance reports through an administrative dashboard.

III. AI Recognition Process

The uploaded classroom image is preprocessed using OpenCV techniques including denoising, color correction, histogram equalization, and contrast enhancement. RetinaFace detects faces and facial landmarks, after which detected faces are geometrically aligned to reduce variations caused by head orientation and camera angle.

Facial embeddings are generated using ArcFace or FaceNet and compared with encrypted student facial templates using cosine similarity. If the similarity score exceeds a predefined confidence threshold, the student is marked as present; otherwise, the result is flagged for manual review.

IV. System Architecture and Technologies

CampusEYE consists of Student Registration, Dataset Preparation, Face Detection, Face Recognition, Attendance Verification, and Dashboard modules. Multiple facial images under different lighting conditions are collected during registration, while normalization and augmentation improve dataset quality.

The backend uses Python and Flask, with OpenCV for preprocessing and ArcFace for facial recognition. FAISS performs efficient embedding similarity search, while MySQL stores student and attendance data. React is used for the frontend, with optional Flutter-based mobile access. REST APIs and Docker support portable cloud or on-premise deployment.

V. Security, Privacy and Benefits

Facial embeddings are protected using AES-256 encryption, while TLS secures client-server communication. Role-based access control, audit logs, and student consent support secure and transparent data management. CampusEYE reduces attendance collection time, minimizes human error, prevents proxy attendance, supports simultaneous student recognition, and enables attendance analytics and LMS integration.

VI. Conclusion

CampusEYE demonstrates how AI and facial recognition can modernize classroom attendance through a single classroom photograph. Its modular architecture and scalable deployment model provide an efficient and secure attendance solution. Future work will explore real-time tracking, mask-aware recognition, improved anti-spoofing, and integration with additional campus management systems.